

WEEK 02 ASSIGNMENT

RED TEAM TASK

Network Information

Q1. Using the **ipconfig /all** command, identify your computer's IPv4 Address, MAC Address, and Default Gateway. Explain the purpose of each and attach Screenshots of your results.

IPv4 Address: (192.168.118.1)

- Purpose: This is your computer's unique local identity card on your home or office network. It allows other local devices to find and talk to your computer.
- **IPv4 Address : 192.168.118.1(Preferred)**

MAC Address (Physical Address): (00-50-56-C0-00-08)

- Purpose: This is a permanent, physical hardware ID burned into your computer's network card by the manufacturer. It never changes, no matter what network you connect to.
- **Physical Address : 00-50-56-C0-00-08**

Default Gateway: (Example: 192.168.1.1)

- Purpose: This is the IP address of your router. It acts as the "exit door" your computer uses to send data out to the internet.
- **Default Gateway :**

Basic Networking Commands

Q2. Execute the following networking commands on your computer. Attach screenshots of the output and write 1–2 lines explaining the purpose of each command.

ipconfig /all

```
Connection-specific DNS Suffix . :  
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1  
Physical Address. . . . . : 00-50-56-C0-00-01  
DHCP Enabled. . . . . : No  
Autoconfiguration Enabled . . . : Yes  
Link-local IPv6 Address . . . . : fe80::5e8c:98c5:715b:3619%10(Preferred)  
IPv4 Address. . . . . : 192.168.118.1(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
DHCPv6 IAID . . . . . : 536891478  
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-7D-0A-48-44-8A-5B-A6-D1-2C  
NetBIOS over Tcpip. . . . . : Enabled
```

Ethernet adapter VMware Network Adapter VMnet8:

```
Connection-specific DNS Suffix . :  
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8  
Physical Address. . . . . : 00-50-56-C0-00-08  
DHCP Enabled. . . . . : No  
Autoconfiguration Enabled . . . : Yes  
Link-local IPv6 Address . . . . : fe80::1faf:67dd:b94:45c3%2(Preferred)  
IPv4 Address. . . . . : 192.168.30.1(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
DHCPv6 IAID . . . . . : 570445910  
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-7D-0A-48-44-8A-5B-A6-D1-2C  
NetBIOS over Tcpip. . . . . : Enabled
```

Explanation: Shows all your computer's current network settings, including your IP address, MAC address, and DNS servers.

ping google.com

```
C:\Users\E Z Z O>ping google.com

Pinging google.com [142.250.202.238] with 32 bytes of data:
Reply from 142.250.202.238: bytes=32 time=64ms TTL=115
Reply from 142.250.202.238: bytes=32 time=66ms TTL=115
Reply from 142.250.202.238: bytes=32 time=71ms TTL=115
Reply from 142.250.202.238: bytes=32 time=67ms TTL=115

Ping statistics for 142.250.202.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 71ms, Average = 67ms
```

Explanation: Tests if your computer can successfully connect to Google's server and measures how fast the connection is in milliseconds.

tracert google.com

```
C:\Users\E Z Z O>tracert google.com

Tracing route to google.com [142.250.202.238]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    192.168.100.1
  2   4 ms     2 ms     2 ms     11.200.77.210
  3  25 ms    25 ms    25 ms    59.103.245.57
  4  25 ms    24 ms    24 ms    10.253.12.46
  5  39 ms    39 ms    *        10.253.4.38
  6  *        *        *        Request timed out.
  7  43 ms    41 ms    *        72.14.212.112
  8  39 ms    38 ms    38 ms    172.253.51.55
  9  39 ms    39 ms    39 ms    192.178.98.162
 10  41 ms    41 ms    40 ms    142.251.77.211
 11  43 ms    44 ms    43 ms    192.178.105.71
 12  44 ms    42 ms    42 ms    192.178.96.3
 13  44 ms    43 ms    *        lcmcta-ad-in-f14.1e100.net [142.250.202.238]
 14  43 ms    44 ms    44 ms    lcmcta-ad-in-f14.1e100.net [142.250.202.238]

Trace complete.
```

Explanation: Maps out the exact path (the "hops" or routers) your data travels through to get from your computer to Google.

nslookup google.com

```
C:\Users\E Z Z O>nslookup google.com
Server: dns.google
Address: 8.8.8.8

DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
Non-authoritative answer:
Name:    google.com
Addresses: 2a00:1450:4018:812::200e
          142.250.202.238
```

Explanation: Acts like a phonebook look-up and it converts the human-readable website name (google.com) into its actual numerical IP address.

arp -a

```
C:\Users\E Z Z O>arp -a

Interface: 192.168.30.1 --- 0x2
    Internet Address      Physical Address      Type
    192.168.30.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static

Interface: 192.168.100.31 --- 0x6
    Internet Address      Physical Address      Type
    192.168.100.1         a0-36-78-f9-13-44    dynamic
    192.168.100.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.118.1 --- 0xa
    Internet Address      Physical Address      Type
    192.168.118.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
```

Explanation: Shows a list of all the local physical hardware addresses (MAC addresses) mapped to IP addresses on your immediate local network.

Router vs. Switch

Q3. Compare a Router and a Switch by creating a table with at least six comparison points, including their functions, OSI layer, addressing, and common use cases.

Comparison Feature	Switch	Router
Primary Function	Connects multiple devices together within the <i>same</i> local network.	Connects multiple <i>different</i> networks together (e.g., your home network to the internet).
OSI Layer	Operates at Layer 2 (Data Link Layer).	Operates at Layer 3 (Network Layer).
Addressing Used	Uses MAC Addresses to move data.	Uses IP Addresses to move data.
Data Format	Deals with data in units called Frames .	Deals with data in units called Packets .
Intelligence	Only knows about devices directly plugged into it.	Smart enough to find the quickest path across the internet to route data.
Common Use Case	Used to connect computers, printers, and consoles inside an office or room.	Used to connect your home or office to your Internet Service Provider (ISP).

IPv4 vs. IPv6

Q4. Compare IPv4 and IPv6 by listing at least six key differences in a table. Include one valid example of an IPv4 address and one valid example of an IPv6 address.

Feature	IPv4	IPv6
Address Size	32-bit address.	128-bit address.
Format	Written in numbers separated by dots (Decimal).	Written in numbers and letters separated by colons (Hexadecimal).
Number of Addresses	About 4.3 billion (Running out globally).	Practically infinite
Security	Security features (IPsec) are optional.	Built-in native security features (IPsec is mandatory).
Configuration	Usually requires manual setup or a DHCP server.	Supports auto-configuration (devices configure themselves).
Valid Example	192.168.1.1	2001:0db8:85a3:0000:0000:8a2e:0370:7334

OSI Model

Q5. List all seven layers of the OSI Model in the correct order. For each layer, write:
Its primary function One commonly used protocol One networking device
associated with that layer

1. Application Layer

- **Primary Function:** Provides network services directly to user applications (like web browsers or email clients).
- **Protocol:** HTTP / HTTPS
- **Device:** Gateway / Firewall

2. Presentation Layer

- **Primary Function:** Translates, encrypts, and compresses data so the application layer can understand it.
- **Protocol:** SSL / TLS
- **Device:** Gateway / PC (handled in software)

3. Session Layer

- **Primary Function:** Opens, manages, and closes communication sessions between two devices.
- **Protocol:** NetBIOS
- **Device:** Gateway / PC (handled in software)

4. Transport Layer

- **Primary Function:** Breaks data into pieces and ensures it travels reliably and accurately from end to end.
- **Protocol:** TCP / UDP
- **Device:** Layer 4 Firewall

5. Network Layer

- **Primary Function:** Finds the best physical path for data to travel across different networks using logical addresses.

- **Protocol:** IP (IPv4 / IPv6)
- **Device:** Router

6. Data Link Layer

- **Primary Function:** Handles node-to-node data transfer on the same network and checks for physical transmission errors.
- **Protocol:** Ethernet
- **Device:** Network Switch

7. Physical Layer

- **Primary Function:** Transmits raw bits (1s and 0s) over physical mediums like cables or radio waves.
- **Protocol:** Wi-Fi (802.11) / DSL
- **Device:** Hub / Network Cables / Repeater

